

Design an Optimal ANFIS Controller using Bee Colony Optimization for Trajectory Tracking of a Quadrotor UAV

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Abstract This paper presents a new neuro-fuzzy bee colony optimization method to find the optimal distribution of the membership functions (MFs) in the design of neuro-fuzzy controllers for complex nonlinear systems. The paper proposes an intelligent controller based on adaptive network-based fuzzy inference system (ANFIS) and bee colony optimization (BCO) algorithm to govern the behavior of a three-degree-of-freedom quadrotor unmanned aerial vehicle. The quadrotor was chosen due to its simple mechanical structure; nevertheless, these types of aircraft are highly nonlinear. Intelligent control such as fuzzy logic is a suitable choice for controlling nonlinear systems. The ANFIS controller is used to reproduce the desired trajectory of the quadrotor in 2D vertical plane, and the BCO algorithm aims to dynamically find the optimal distribution of the MFs at the input and output ends of the ANFIS in order to reduce learning errors and improve the quality of

the controller. To evaluate the performance of the proposed BCO-tuned ANFIS controller, a comparison between the proposed ANFIS-BCO controller and other controller's performance such as ANFIS and proportional-integral-derivative controllers is illustrated using the same system. The comparison of results shows that the proposed ANFIS-BCO method outperforms the other methods already developed in the same study.

Keywords Unmanned aerial vehicle · Optimal control · Adaptive neuro-fuzzy inference system · Bee colony algorithm

Introduction

Development of unmanned aerial vehicles (UAVs) has become the most important research areas in the field of autonomous aeronautical control. UAVs technology has continuously been evolving with exceptional growth over the last years [1], leading to the emergence of a large number of services offered and potential applications. Drones are not meant to only serve with military purposes [2], but have also become widely used in civilian and industrial domain such as logistics and transportation [3–5], photography and filmmaking [6], safety and security [7], mapping [8], agriculture [9, 10], monitoring [11–14], surveillance [15–17], architecture [18–20] and others applications. They were originally used for missions tedious or too dangerous for humans and all this due to their ease of deployment, low maintenance cost [21], high mobility and hovering capability [22]. In general, UAVs are preferred for their ability to stabilize at a particular position and altitude [23], to fly at various speeds [24–26], to hover in a stationary position over a target [22, 27] and

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to perform all these maneuvers in close proximity to obstacles [28, 29].

Generally, there are two steps for designing controllers, first the selection of a specific structure and, second, the computation of suitable controller parameters. Manual tuning methods are iterative and time-consuming, and if used on hardware, they can cause damage. Rule-based methods also have serious limitations: They do not support certain types of plant models, such as unstable plants, high-order plants, or plants with little or no time delay. On another side, these methods require numerical computations in order to get the best possible controller parameters. For these reasons, considerable research effort has been elaborated in heuristic optimization methods to guarantee optimal or near optimal results.

The controller design process is to find the optimal values for controller parameters from the search space that minimizes the objective function. The performance of the heuristic algorithm-based controller tuning mainly depends on the objective function, which guides the optimization search to obtain appropriate value for the controller parameters.

A considerable number of meta-heuristic search methods are proposed by the researchers to find optimal solutions for a class of engineering optimization problems. The most commonly used methods in particular for controllers optimization that can be found in the literature are as follows: bee colony optimization (BCO) [30, 31], ant colony optimization (ACO) [32–39], particle swarm optimization (PSO) [40, 41], bacterial foraging optimization (BFO) [42–44] and genetic algorithm (GA) [45–47].

In this paper, a robust nonlinear controller based on adaptive network-based fuzzy inference system (ANFIS) and bee colony optimization (BCO) algorithm is used to govern the behavior of a three-degree-of-freedom quadrotor unmanned aerial vehicle. The BCO algorithm aims to facilitate convergence towards the optimal ANFIS parameters to reduce learning errors and improve the quality of the controller.

This paper is organized as follows. The description of the UAV model and the problem formulation are given in Sect. 2. ANFIS system with its architecture and learning algorithm are introduced in Sect. 3. The purpose of Sect. 4 is to present the BCO algorithm that will later be used for the optimization of ANFIS parameters. Section 5 details the proposed control design and strategy. In Sect. 6, comparisons and numerical simulations results are given to demonstrate the effectiveness of the proposed controller. Section 7 gives the conclusions.

Model Description and Problem Formulation

Description of the Quadrotor Model

It is primordial to introduce the reference coordinates to describe the full structure of the quadrotor. The quadrotor body is defined in y, z axes frame. As shown in Fig. 1, the origin is located in the center of the 3DOF quadrotor. The Euler angle ϕ represents the orientation, defining the roll angle about the horizontal axis.

Therefore, the quadrotor is modeled to fly in two dimensions $y - z$ plane with a roll angle ϕ as shown in Fig. 1.

The state of the quadrotor is therefore $[y, z, \phi]^T$ and there are two inputs u_1 and u_2 , which represent the thrust and the moment about the x -axis, respectively (Fig. 1).

Equation of Motion

Since our drone is modeled in two dimensions, we have y and z plane and a roll angle ϕ as seen before. The equations describing the movement are written as follows:

$$\ddot{y} = \frac{u_1}{m} \sin(\phi). \quad (1)$$

$$\ddot{z} = g - \frac{u_1}{m} \cos(\phi). \quad (2)$$

$$\ddot{\phi} = \frac{u_2}{I_{xx}} \quad (3)$$

where m , g and I_{xx} are the mass, the intensity of gravity and the moment of inertia, respectively.

The equations of motion can be written as follows:

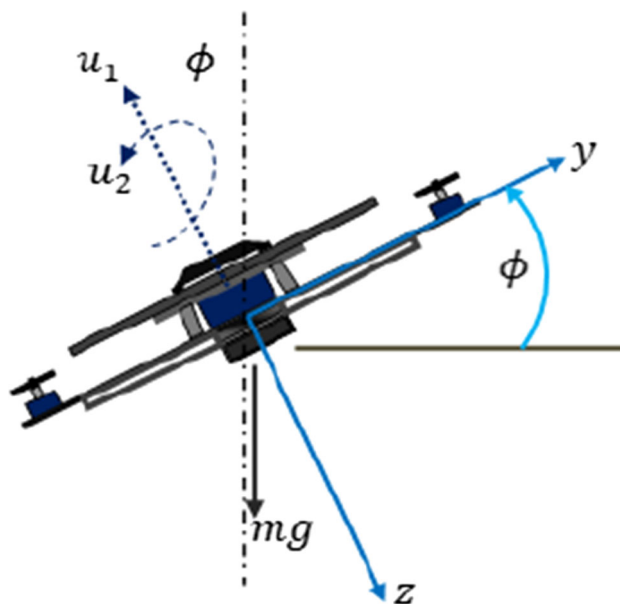


Fig. 1 Quadrotor configuration design

$$\begin{bmatrix} \ddot{y} \\ \ddot{z} \\ \ddot{\phi} \end{bmatrix} = \begin{bmatrix} 0 \\ g \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{1}{m}\sin\phi & 0 \\ -\frac{1}{m}\cos\phi & 0 \\ 0 & \frac{1}{I_{zz}} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}. \quad (4)$$

The state space description of the quadrotor is as follows:

$$\dot{x} = \begin{bmatrix} \dot{y} \\ \dot{z} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} y \\ z \\ \phi \\ \dot{y} \\ \dot{z} \\ \dot{\phi} \end{bmatrix}. \quad (5)$$

So, the first derivative of the state vector is expressed in Eq. (6). The first three parameters of the x vector represent velocities and the last three parameters represent accelerations.

$$\dot{x} = \begin{bmatrix} \dot{y} \\ \dot{z} \\ \dot{\phi} \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m}\sin\phi & 0 \\ -\frac{1}{m}\cos\phi & 0 \\ 0 & \frac{1}{I_{zz}} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}. \quad (6)$$

The vector $[u_1, u_2]$ is the input signals that can drive the dynamical system; by specifying the properties of u_1 and u_2 , we can change the state of the quadrotor.

From equation of motion in (1) and (2), it is seen that the studied system is nonlinear, and this nonlinearity comes from the existence of $\cos(\phi)$ and $\sin(\phi)$ terms. In those conditions, we need to perform a linearization; let's consider that the motion equations have a hover configuration.

Equilibrium Hover Configuration

Only when the thrust vector u_1 and the gravity vector are opposed, the hover configuration will be in equilibrium. The system is linearized near the hover point, where the properties are:

$$y_0, z_0, \phi_0 = 0, u_{1,0} = mg, u_{2,0} = 0,$$

Linearized Dynamics Model

We know that the $\sin(\phi)$ and $\cos(\phi)$ are the two sources of the nonlinearity, so when ϕ is close to zero, $\sin(\phi)$ behaves basically like ϕ and $\cos(\phi)$ becomes nearly constant.

$$\ddot{y} = \frac{mg}{m} \cdot \sin\Delta\phi \rightarrow g\phi \quad (7)$$

$$\ddot{z} = g - \frac{u_1}{m} \cdot \cos\Delta\phi \rightarrow g - \frac{u_1}{m}. \quad (8)$$

$$\ddot{\phi} = \frac{u_2}{I_{xx}} \quad (9)$$

If we replace $\sin(\phi)$ by ϕ and $\cos(\phi)$ by 1, we get the linear equations as seen below,

$$\ddot{y} = g\phi. \quad (10)$$

$$\ddot{z} = g - \frac{u_1}{m}. \quad (11)$$

$$\ddot{\phi} = \frac{u_2}{I_{xx}}. \quad (12)$$

ANFIS System

The proposed neuro-fuzzy network is a five-layer architecture that includes the elements of a fuzzy Sugeno-type system [48–50]. Looking at Fig. 2, let us explain how the network works layer by layer.

Layer 1 (Fuzzification): This layer contains adaptive nodes. The outputs are the fuzzy membership grade of the inputs, which are given by:

$$O_i^1 = \mu_{A_i}(x), \quad i = 1, 2. \quad (13)$$

$$O_j^1 = \mu_{B_j}(y), \quad j = 1, 2. \quad (14)$$

where μ_{A_i}, μ_{B_j} denote the membership degrees obtained from this layer.

Fuzzifying the inputs is conducted by membership function (MF) such as piecewise linear, triangular, trapezoidal, Gaussian and singleton. Among the abovementioned MFs, in this paper the Gaussian function has been used for its smooth and concise notation. Therefore, $\mu_{A_i}(x)$ is expressed by;

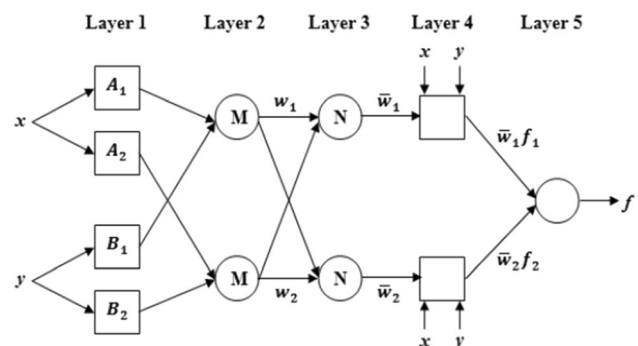


Fig. 2 The equivalent typical ANFIS architecture

$$\text{Triangular : } \mu_{A_i}(x) = \max\left(\min\left(\frac{x-a_i}{b_i-a_i}, \frac{c_i-x}{c_i-b_i}\right), 0\right), i = 1, 2. \quad (15)$$

$$\text{Trapezoidal : } \mu_{A_i}(x) = \max\left(\min\left(\frac{x-a_i}{b_i-a_i}, 1, \frac{d_i-x}{d_i-c_i}\right), 0\right), i = 1, 2. \quad (16)$$

$$\text{Gaussian : } \mu_{A_i}(x) = \exp\left(-\frac{(x-c_i)^2}{\sigma_i^2}\right), i = 1, 2. \quad (17)$$

where a_i , b_i , c_i and σ_i are the premise parameters.

Layer 2 (Weighting of fuzzy rules): The symbol M shows every fixed node in this layer. This layer calculates the firing strength w_k by using membership values computed in fuzzification layer, and the outputs are computed as follows:

$$O_k^2 = w_k = \mu_{A_i}(x) * \mu_{B_j}(y), i, j = 1, 2 \quad (18)$$

Layer 3 (Normalization): Every node is fixed node and called by N . Each node obtains the normalization by calculating the ratio of the k th rule's firing strength (truth values) to the sum of all rules firing strength. The output O_k^3 at this step is given by:

$$O_k^3 = \overline{w}_k = \frac{w_k}{\sum w_i} = \frac{w_k}{w_1 + w_2}, \quad k = 1, 2. \quad (19)$$

Layer 4 (Defuzzification): Weighted consequent values of rules are calculated in each node of this layer as given in Eq. (20).

$$O_k^4 = \overline{w}_k f_k = \overline{w}_k(p_k x + q_k y + r_k), \quad k = 1, 2. \quad (20)$$

where w_k represents the output of the third layer and $\{p_k, q_k, r_k\}$ are consequent.

Layer 5 (Summation): The actual output is obtained by summing the outputs of all incoming signals that are coming from the defuzzification layer to produce the overall ANFIS output.

$$O^5 = \sum_{k=1}^2 \overline{w}_k f_k = \frac{\sum_{k=1}^2 w_k \cdot f_k}{w_1 + w_2}. \quad (21)$$

Bee Colony Optimization Algorithm

Bee colony optimization (BCO) is an optimization technique which simulates the intelligent foraging behavior of honey bees in nature. BCO is a new meta-heuristic designed to deal with hard combinatorial optimization problems. It is biologically inspired method that explores collective intelligence applied by the honey bees during food search (nectar collecting process). This optimization

technique has been proposed by Marco Karaboga in 2005 [51]. A set of honey bees is called swarm which can successfully accomplish tasks through social cooperation. In the BCO algorithm, there are three types of bees: employed bees, onlooker bees and scout bees. The employed bees search food around the food source in their memory; meanwhile, they share the information of these food sources to the onlooker bees. The onlooker bees tend to select good food sources from those found by the employed bees. The food source that has higher quality (fitness) will have a large chance to be selected by the onlooker bees than the one of lower quality.

In the ABC algorithm, the first half of the swarm consists of employed bees, and the second half constitutes the onlooker bees. The number of employed bees or the onlooker bees is equal to the number of solutions in the swarm [51].

Mathematical Model of BCO

The BCO generates a randomly distributed initial population of SN solutions (food sources), where SN denotes the swarm size. Let $X_i = \{x_{i,1}, x_{i,2}, \dots, x_{i,D}\}$ represent the i th solution in the swarm, where D is the dimension size. Each employed bee X_i generates a new candidate solution V_i in the neighborhood of its present position as follows:

$$v_{i,j} = x_{i,j} + \varnothing_{i,j} \cdot (x_{i,j} - x_{k,j}). \quad (22)$$

where X_k is a randomly selected candidate solution $i \neq k$, j is a random dimension index selected from the set $\{1, 2, \dots, D\}$, and $\varnothing_{i,j}$ is a random number within $[-1, 1]$. Once the new candidate solution V_i is generated, a greedy selection is used. If the fitness value of V_i is better than that of its parent X_i , then update X_i with V_i ; otherwise keep unchangeable.

After all employed bees complete the search process, they share the information of their food sources with the onlooker bees through waggle dances. An onlooker bee evaluates the nectar information taken from all employed bees and chooses a food source with a probability related to its nectar amount. This probabilistic selection is really a roulette wheel selection mechanism which is described as follows:

$$P_i = \frac{\text{fit}(x_i)}{\sum_{j=1}^{SN} \text{fit}(x_j)}. \quad (23)$$

where $\text{fit}(x_i)$ is the fitness value of the i th solution in the swarm. As seen, the better the solution i , the higher the probability of the i th food source selected.

If a position cannot be improved over a predefined number of cycles, then the food source is abandoned. Assume that the abandoned source is X_i , then the scout bee

discovers a new food source to be replaced with X_i as follows:

$$x_{i,j} = lb_j + rand(0, 1) \cdot (ub_j - lb_j). \quad (24)$$

where $rand(0, 1)$ is a random number within $[0, 1]$ based on a normal distribution and lb_j, ub_j are lower and upper boundaries of j th dimension, respectively.

Controller Design

The auto-tuning (self-optimization) of ANFIS parameters is specifically to find the optimal design of the MFs in the neuro-fuzzy controller by a bee colony algorithm leads to automatically adjusting the fuzzy rules because the two parts (membership functions and fuzzy rules) cannot be dissociated. Moreover, the optimization by the BCO, whose sole objective is the improvement of a numerical criterion, often leads to the presence of more precise fuzzy rules at the end of the optimization process which leads to better results. It is within this framework that our motives and interests lie in the design of neuro-fuzzy controllers by bee colony algorithms.

In our study, an ANFIS controller applied to a totally autonomous UAV tracking trajectory, modeled to fly in 2 dimensions (yz) plane with an angle ϕ which is the roll angle; both y and z equations that describe the trajectory and the roll angle ϕ are considered as an objective function. To improve controller, an optimization method based on the evolutionary BCO algorithms is proposed to obtain the optimal parameters leading to the optimal or ideal trajectory and therefore to completion of the specified movement resets the control.

The trajectory to follow is defined by the vector R_T , which consisting of two elements $\{y(t), z(t)\}$ as shown in Fig. 3.

Given the reference trajectory

$$R_T(t) = \begin{bmatrix} y(t) \\ z(t) \end{bmatrix}. \quad (25)$$

And the measured trajectory is defined by the vector R_C

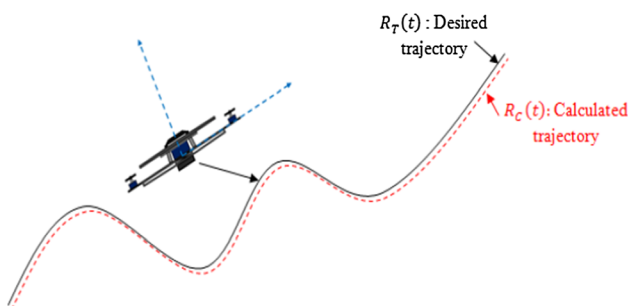


Fig. 3 Quadrotor trajectory tracking in 2D plane

$$R_C(t) = \begin{bmatrix} y_C(t) \\ z_C(t) \end{bmatrix}. \quad (26)$$

where $y_C(t)$ and $z_C(t)$ are the calculated parameters by the controller.

For each parameter defining the trajectory, we consider a controller with two inputs: the error $e(t)$ and its variation $\Delta e(t)$ and one output: the setpoint change $\Delta u(t)$, which allows to adjust online the command $u(t)$, applied to the system (Fig. 4). The fuzzy rules constituting the base of the controller, in this case, have two premises.

The proposed ANFIS controller has been used to represent the objective function, which allows optimal control of the intelligent UAV, and a BCO algorithm to obtain the optimal objective value. The optimization process of the ANFIS controller by BCO is illustrated by block schema given in Fig. 4.

The population of agents (artificial bees) consisting of B bees collaboratively searches for the optimal solution. To optimize the parameters of the ANFIS controller by BCO, we first need to create the controller design. This hybridization can be represented by a graph as given in Fig. 5 which is composed of nodes (circles) and edge (connections). Each node indicates the following solution that an artificial bee in the search space can choose as a solution to the problem. All the values of the parameters ($iMF1_i, iMF2_i, oMF_i$) can be considered as three vectors, and each parameter contains three parameters that define the membership functions as shown above, which $iMF1_i, iMF2_i$ and oMF_i are the MFs of the first input, the second input and the output, respectively. In order to create a representation in graph form correspond to the problem, these three vectors represent paths between the start node and the end node. To make a complete tour, each bee must visit the three food sources by following path between start node and end node. BCO's objective is to find the optimal tour that has the lowest cost function. This fitness function is as follows.

$$\varphi = \frac{1}{n} \sum_{k=1}^n \sqrt{(y_k - \hat{y}_k)^2} \quad (27)$$

where φ is the cost value, k is the order of the data depending on the sampling time, n is the total number of data, y is the input desired trajectory and $\hat{y}$ is the system output trajectory.

In a neuro-fuzzy controller, each linguistic variable is defined by a set of membership functions of the language terms, and fuzzy rules are applied to linguistic terms. These terms, which qualify a linguistic variable, are defined through MFs.

The membership function is defined by parameters like the triangular one as shown in Fig. 6, and it is defined by

Fig. 4 Optimization ANFIS controller with BCO block diagram

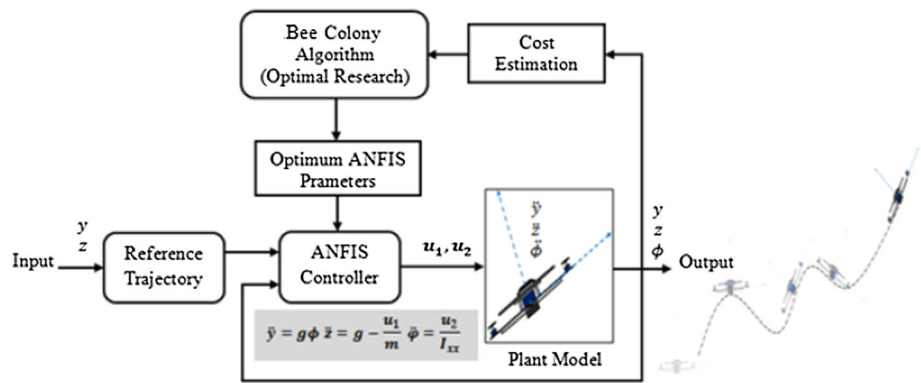


Fig. 5 BCO for ANFIS tuning process

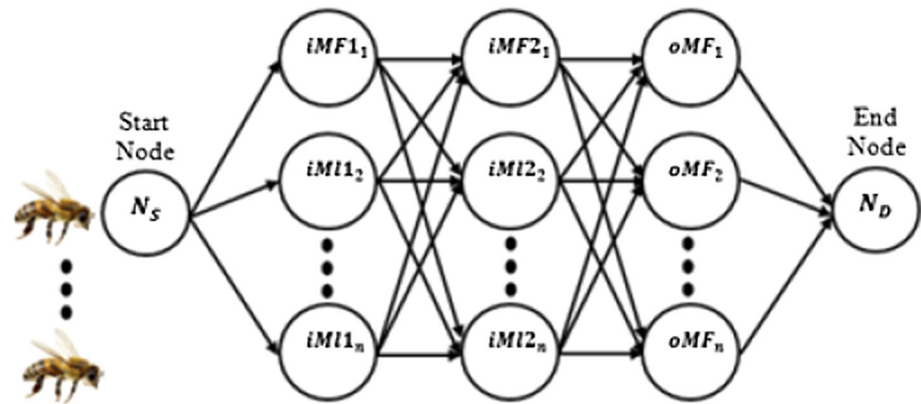
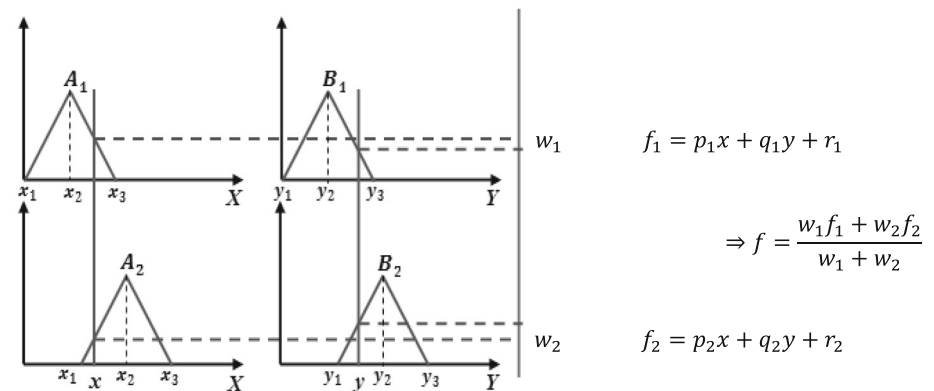


Fig. 6 A two-inputs first-order Sugeno-type model with two rules



three parameters x_1, x_2 and x_3 which take their values in the interval $[a, b]$.

The optimization procedure consists in finding the best adjustment of the parameters of the MFs of each control variable.

In general, the objective of a control system is to minimize the difference $e(t)$ of the system output and a desired (reference) output. This difference may be due either to a set-point change or to disturbances acting on the system. This objective can be defined by several numerical indices. Only the desired behavior can be a paramount parameter to take into account to make a good choice among these indices. In our research work, we have opted for the

minimization of the mean squared error (MSE) and root mean squared error (RMSE).

After selecting the BCO parameters, each bee is allowed to explore and find a full path travel when leaving the hive the bees observe random dances performed by other bees. Then these bees are equipped with an array of movements of the observed dances. This set of moves referred to as preferred path, and it will serve as a guide in the process of foraging. It contains a complete path that was previously explored by its partner who will lead the bee to the destination. During foraging, a bee goes from a node to another node until the destination is reached. ANFIS is optimized by adjusting MF parameters (c, σ) and consequent

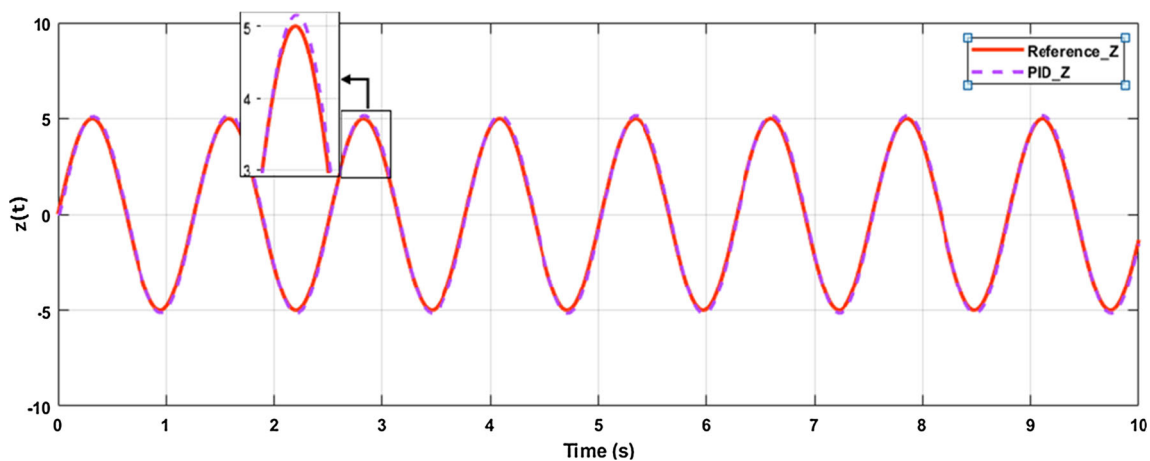
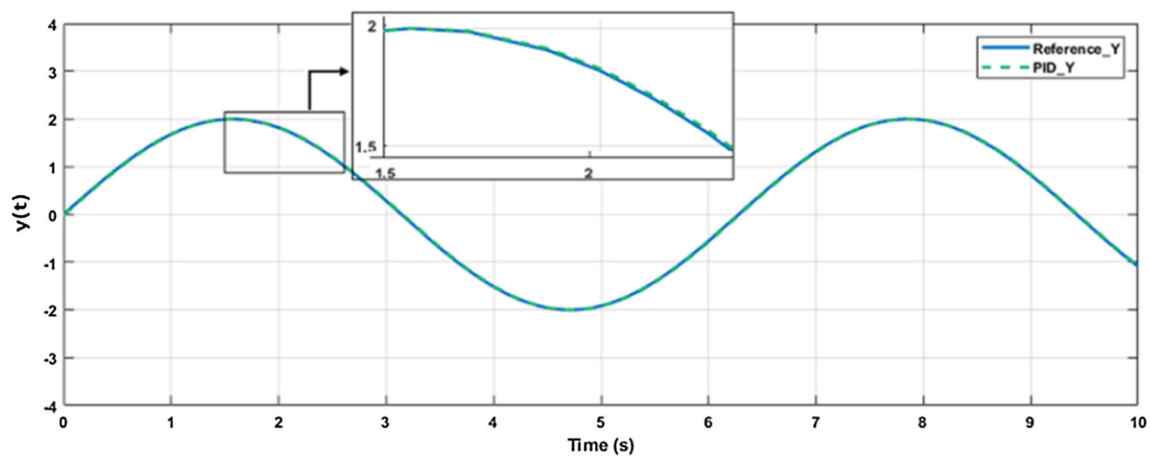
Table 1 Parameters of quadrotor

Symbol	Value
m	0.2 kg
I_{xx}	0.1 kg.m ²
g	9.81 m/s ²

parameter(p, q, r) for approximating the desired output. After initialization, the positions of food sources, bees are repeatedly updated until maximum iterations. In each iteration, Eq. (24) is utilized to search new food sources and compute their nectar. The food sources are chosen based on probability using Eq. (23), and produce new food sources by Eq. (24) in the selected neighborhood. Then, the new neighborhood is exploited for better solutions. In the end of each iteration, the quality of food source is checked by computing nectar amount. A food source is considered

to be exhausted if the number of maximum cycles is reached, or the quality of it cannot be further improved. In this case, bees are sent to search for new food sources randomly using Eq. (24).

The performance of ANFIS depends on how well its structure learning and parameter tuning is performed. The prior refers to the process of obtaining MFs parameters with which ANFIS can approximate the desired output. Whereas, parameter tuning helps in maximizing accuracy of the model. As per modified ANFIS learning algorithm, BCO is employed to update the premise and consequent parameters. Subsequently, the error measure between each actual output and the desired output is calculated using - mean squared error (MSE) and root mean squared error (RMSE). The evolution of the cost function of the BCO is reduced after each iteration until it reaches the optimal controller ANFIS parameters.

**Fig. 7** Measured $z(t)$ trajectory with PID controller**Fig. 8** Measured $y(t)$ trajectory with PID controller

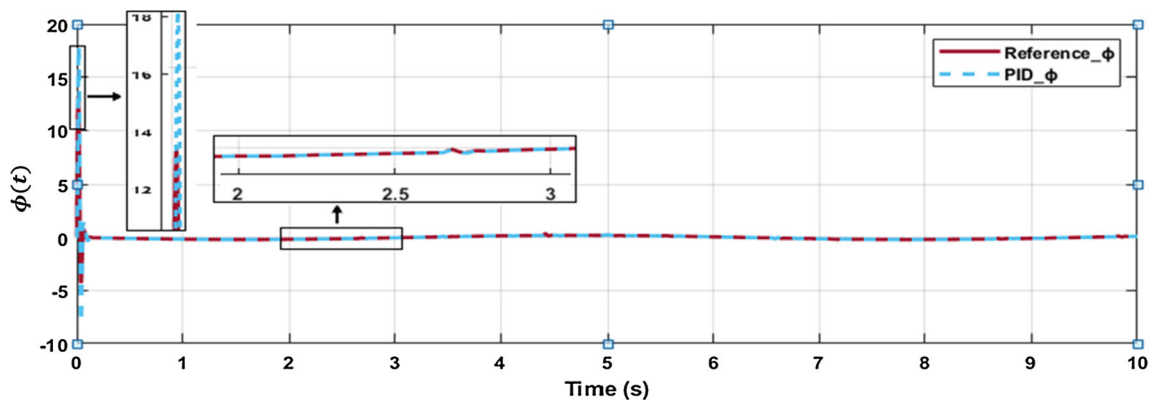


Fig. 9 Measured $\phi(t)$ trajectory with PID controller

Fig. 10 Measured trajectory with PID controller

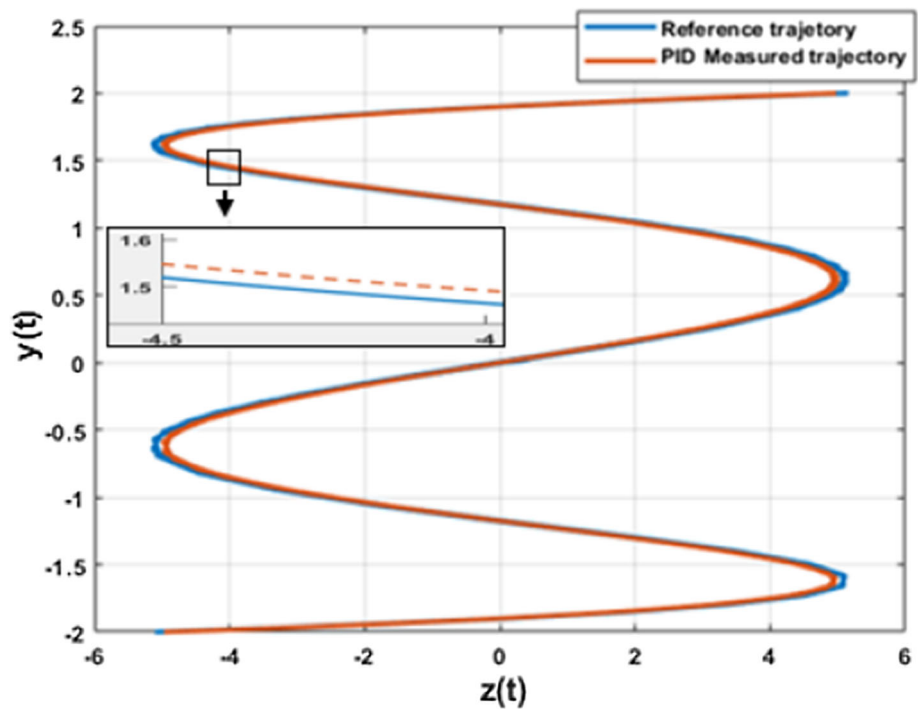
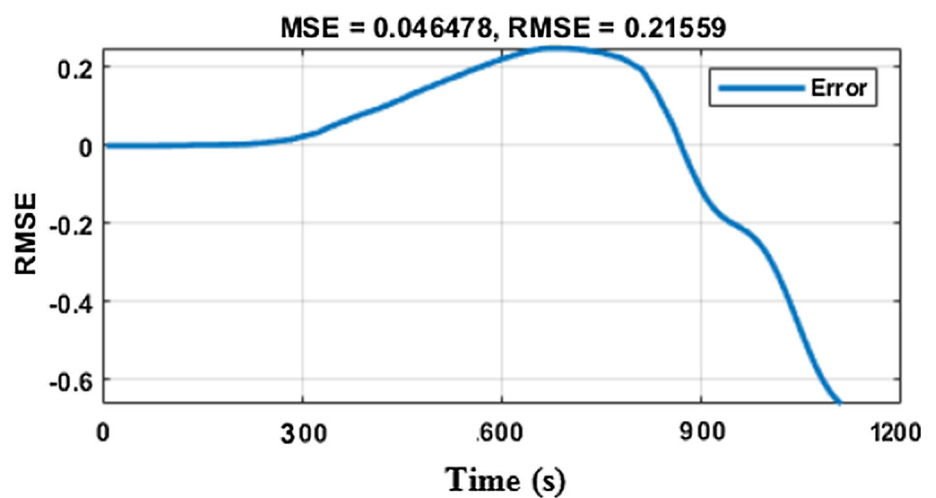


Fig. 11 Trajectory error with PID controller



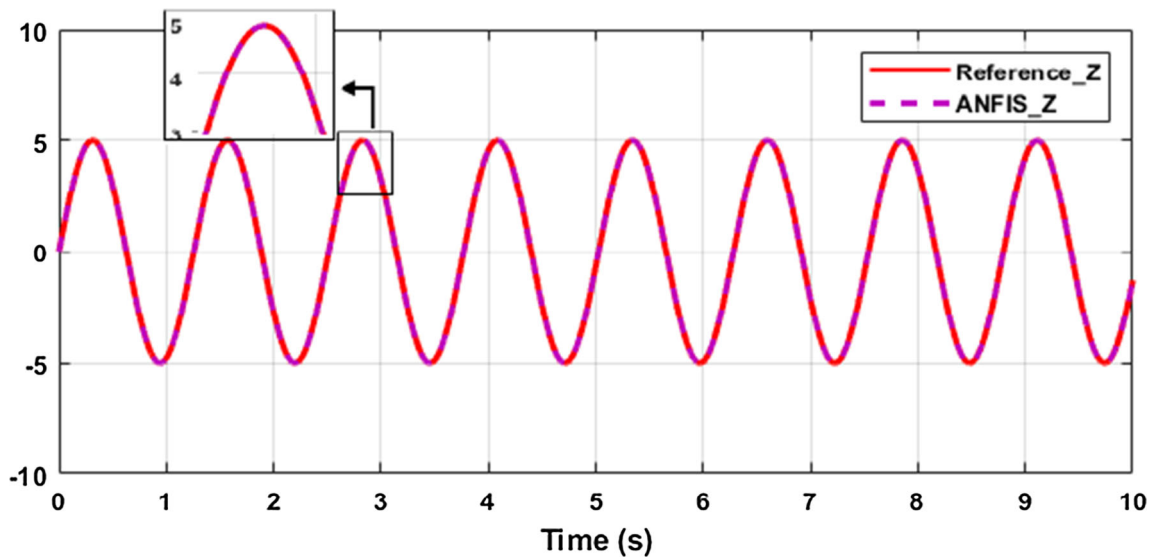


Fig. 12 Measured $z(t)$ trajectory with ANFIS controller

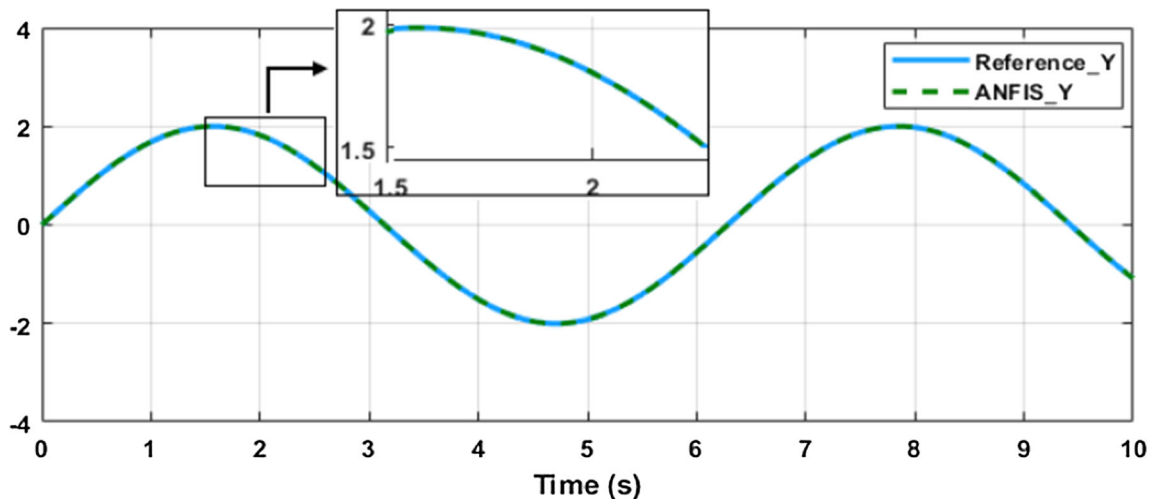


Fig. 13 Measured $y(t)$ trajectory with ANFIS controller

Simulation Results

To validate the proposed approach, we carried out simulations for the adaptive control of a 3-DOF UAV robot moving along a specified trajectory. The model was implemented in Matlab/Simulink programming software. The UAV is commanded to flight following a predefined trajectory as a function of time defined by $y(t)$ and $z(t)$.

The control performance obtained by the proposed control system is compared with those obtained by PID and ANFIS controllers to validate the superior performance of the studied controller.

The simulation results are obtained with a vertical trajectory in 2D space. The desired trajectory input is defined as:

$$y_d(t) = 2\sin(t) \text{ and } z_d(t) = 5\sin(t)$$

The parameters of the quadrotor used in the following simulations are shown in Table 1.

Finally, the three control outputs y , z and ϕ are shown below.

Considering the PID controller, it is clear from simulation results illustrated in Figs. 7, 8, 9 and 10 that the unmanned vehicle follows the predefined trajectory with a delay as strongly seen in Fig. 10 between the reference trajectory (blue line) and the measured one (red line).

The difference between the calculated trajectory and the desired one can be measured by the mean square error as shown in Fig. 11.

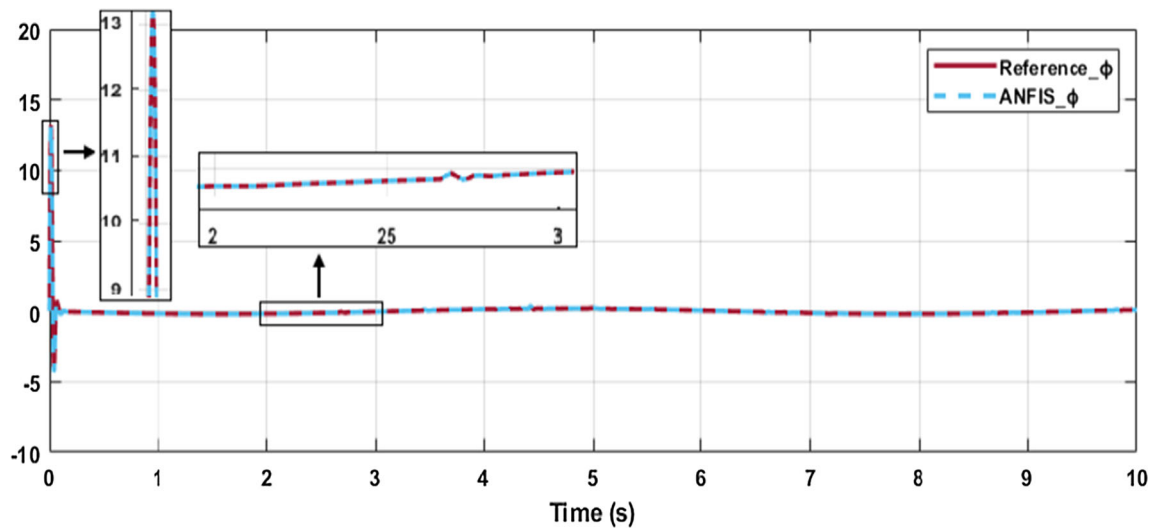


Fig. 14 Measured $\phi(t)$ trajectory with ANFIS controller

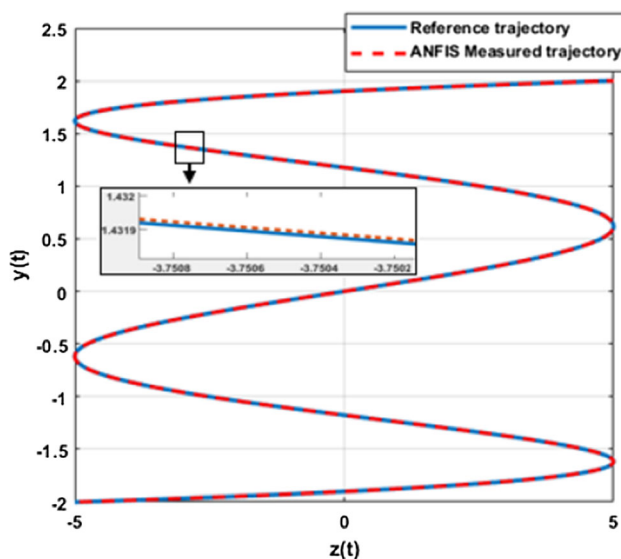


Fig. 15 Measured trajectory with ANFIS controller

As we can see in Fig. 11, the PID controller can reproduce the tracking trajectory with error of $MSE = 4.6478 \times 10^{-2}$ and $RMSE = 0.21559$.

Using the artificial neural fuzzy inference system (ANFIS) controller, results shown in Figs. 12, 13 and 14 indicate that, respectively, the measured $z(t)$ (magenta dotted line), $y(t)$ (green dotted line) and $\phi(t)$ (cyan dotted line) trajectories of the quadrotor track exactly the desired $z(t)$ (red line), $y(t)$ (cyan line) and $\phi(t)$ (brown line) trajectories.

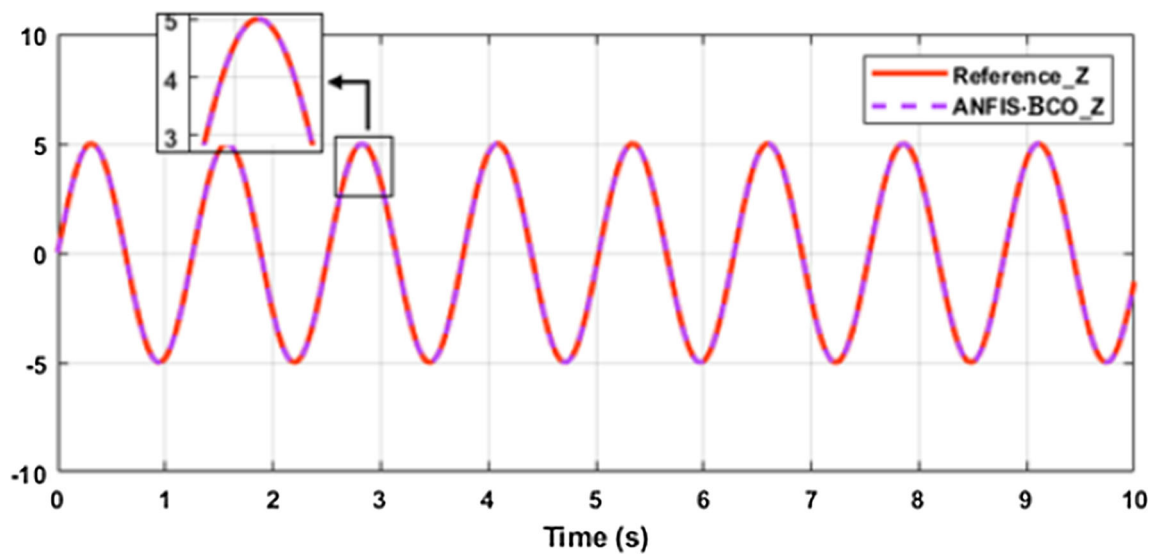
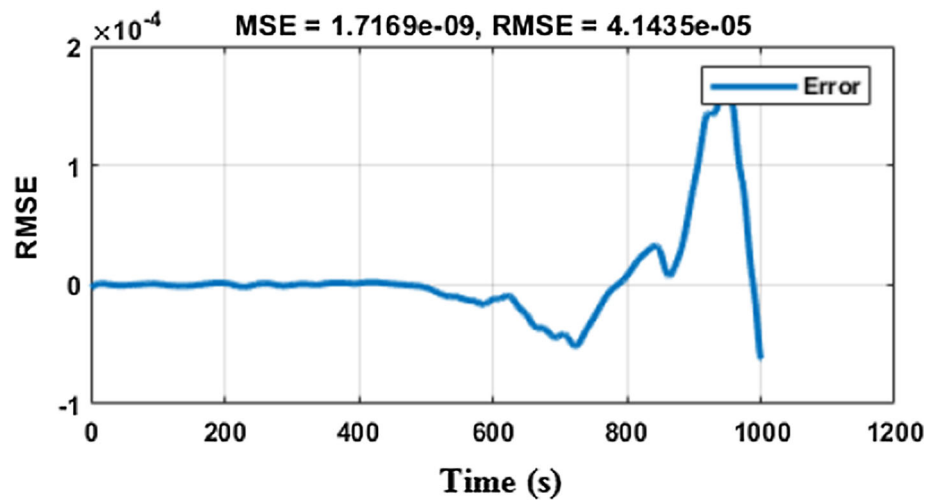
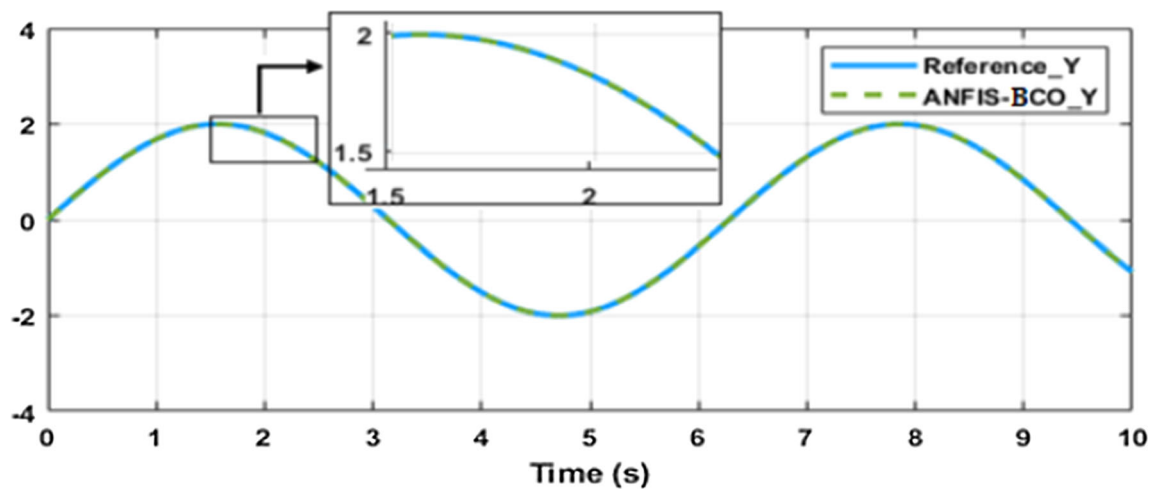
The ANFIS measured ($z(t), y(t)$) trajectory, in Fig. 15 (dotted red line), deviates slightly from the desired (solid blue line) trajectory. The error between the two trajectories shown in Fig. 16 indicates that the calculated mean square error with the ANFIS controller is evaluated at $MSE =$

1.7169×10^{-9} and $RMSE = 4.1435 \times 10^{-5}$, which is too small than the MSE with PID controller which approve the effectiveness of the ANFIS controller over PID controller.

Bioinspired optimization techniques, which are simulated or inspired by biological behaviors of animal, have demonstrated competitive performance on a wide range of optimization problems for this reason we have decided to hybrid the ANFIS controller with the BCO bioinspired algorithm and show its performance by simulations that are shown below.

Figures 17, 18 and 19 show the trajectory in y-axis, z-axis and the quadrotor orientation $\phi(t)$, respectively. From the figures, it is very clear that the measured quadrotor trajectories follow accurately the desired trajectories. The tracking performance is illustrated in Fig. 20, where the measured trajectory (red dotted line) of the quadrotor and the reference trajectory (solid blue line) are shown together. The results obtained show that the performances of the ANFIS-BCO hybrid approach are superior to those of the other models. The validity of the proposed model was proved by mean square error $MSE < 10^{-29}$ and $MSE < 10^{-14}$. The tracking errors of different trajectories ($z(t), y(t)$, and $\phi(t)$) of the quadrotor are illustrated in Figs. 21 to Fig. 26. It can be clearly seen that the trajectory tracking errors converge to zero.

The BCO algorithm provides an improvement, by comparing ANFIS performances with the same ANFIS optimized by BCO, a significant increase in accuracy. A clear improvement of the precision in the trajectory tracking is thus visible. The RMSE used to measure the accuracy of the control model decreases by 10^9 times. The BCO exploration of this search space is performed to identify subsets of more relevant parameters, and accurate by a new membership function distribution, that adapts

Fig. 16 Trajectory Error with ANFIS Controller**Fig. 17** Measured $z(t)$ trajectory with ANFIS-BCO controller**Fig. 18** Measured $y(t)$ trajectory with ANFIS-BCO controller

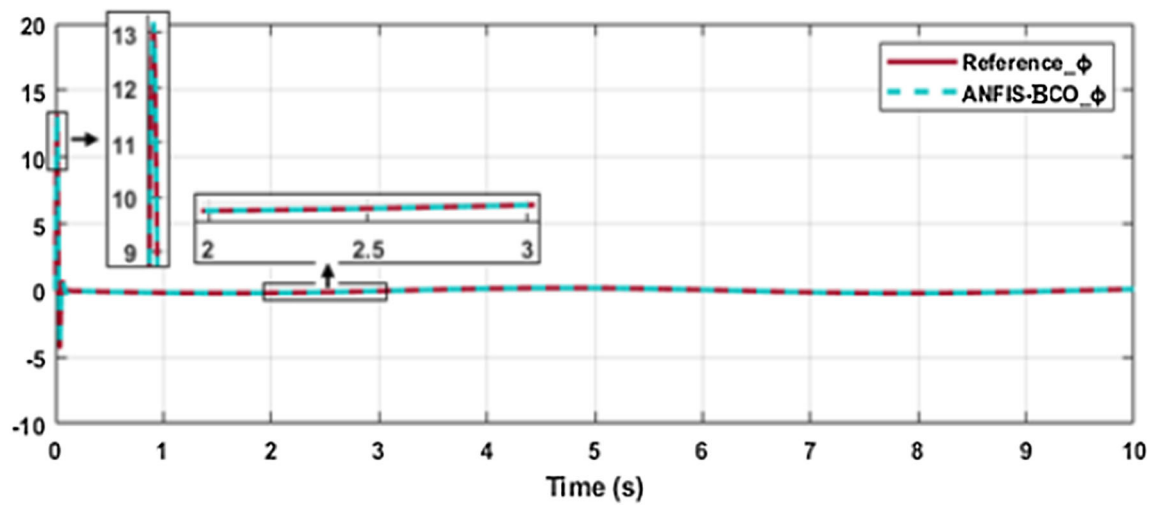


Fig. 19 Measured $\phi(t)$ trajectory with ANFIS-BCO controller

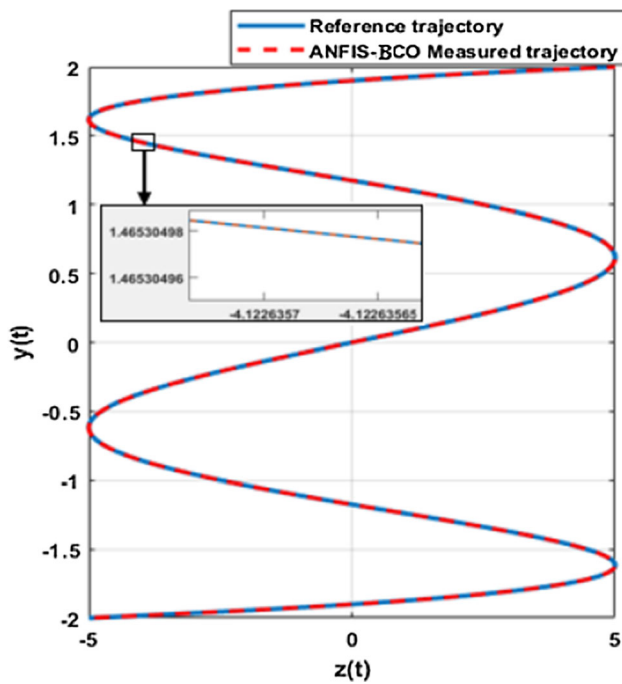


Fig. 20 Measured trajectory with ANFIS-BCO controller

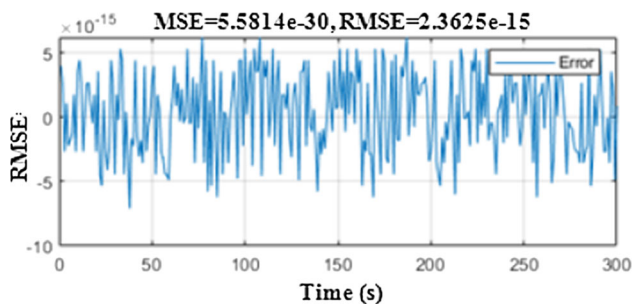


Fig. 21 Error test data (z parameter)

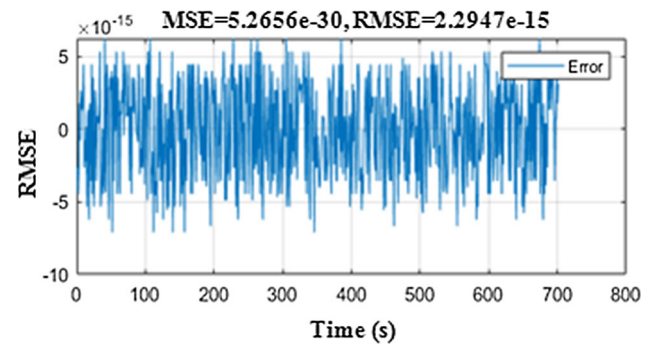


Fig. 22 Error train data (z parameter)

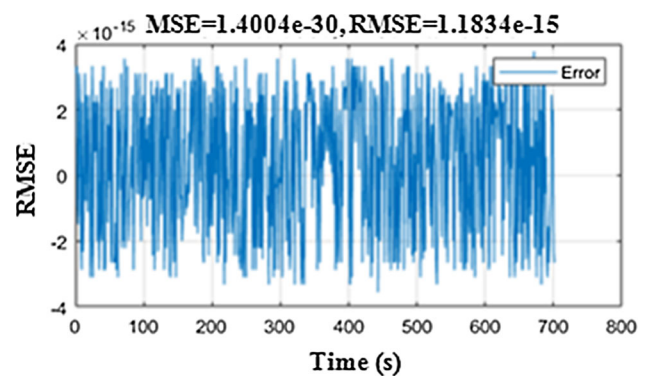
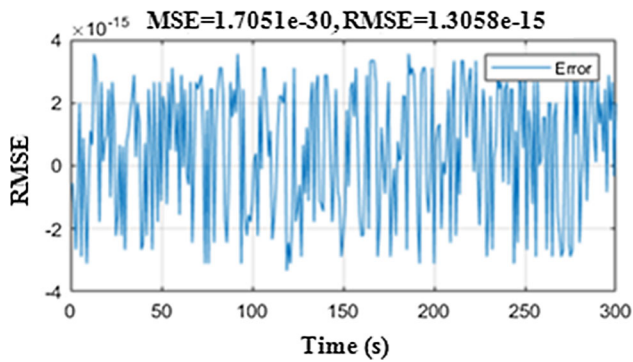
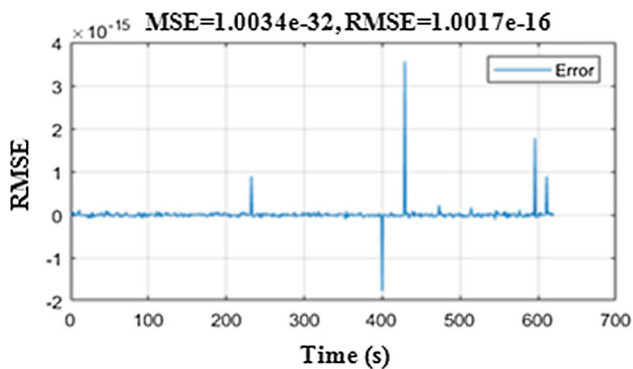
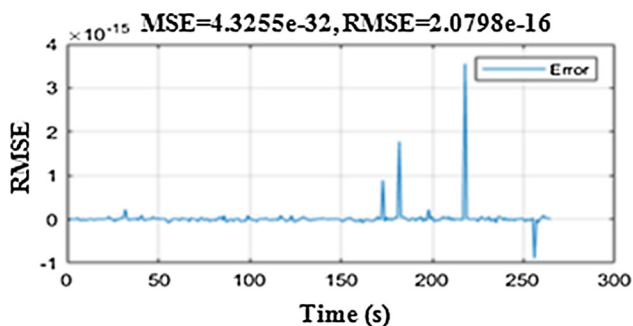


Fig. 24 Error test data (y parameter)

well to each linguistic variable, which leads to minimizing the error between the desired trajectory and the calculated one.

Fig. 23 Error train data (y parameter)Fig. 25 Error train data (ϕ parameter)Fig. 26 Error test data (ϕ parameter)

Conclusion

This paper treats a tuning of neuro-fuzzy controllers method using bee colony optimization (ANFIS-BCO) for the tracking performance of a quadrotor unmanned aerial vehicle. In this paper, we conclude that the BCO algorithm is a good technique of optimization and stabilization for neuro-fuzzy controllers. The BCO algorithm can improve the quality of results and increase the diversity of solutions to a problem. In addition, the BCO algorithm was

implemented to find the optimal distribution of parameters in the design of the ANFIS controller, especially in the fuzzy controllers, thus being able to demonstrate the efficiency of the BCO algorithm as a technique to improve the performances in fuzzy controllers. The simulation results depict that the proposed optimized fuzzy ANFIS-BCO controller performance outperforms the PID and ANFIS controllers. Several reference signals were tested to evaluate the tracking performance where the outcomes demonstrate that the proposed ANFIS-BCO controller was able to minimize the error and bring the quadrotor to the desired trajectory. In addition, RMSE error was reduced significantly using the proposed controller.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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